Cognitive Modeling

Complete Lecture Notes

Week 5: Uncertainty in Estimates and Fitting Multiple Participants

Contents

1 Parameter Estimation Methods 1

1.1 Week 5: Uncertainty in Estimates and Fitting Multiple Participants 1

1.1.1 Key terms 1

1.1.2 Learning outcomes 2

1.1.3 Motivation & big picture 3

1.1.4 Lecture notes 4

1.1.5 Worked example 9

1.1.6 Lab (2 hours) 9

1.1.7 Connection to AI & LLMs 11

1.1.8 Common mistakes & misconceptions 11

1.1.9 Check your understanding 12

1.1.10 Assignment 13

1.1.11 Further reading 13

Chapter 1

Parameter Estimation Methods

1.1 Week 5: Uncertainty in Estimates and Fitting Multiple Participants

Chapter 2: Parameter estimation methods · Lecture 1 h · Lab 2 h Running data: the Week 2–4 forgetting data (how certain are our estimates?) and a new simulated practice experiment with 20 participants (the power law of practice).

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

Table 1.1: Key terms

English	فارسی	توضیح کوتاه
estimate / estimator	برآورد / برآوردگر	روش یا مقدار به‌دست‌آمده برای پارامتر از روی داده
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
power law of practice	قانون توانی تمرین	کاهش توانی زمان واکنش با تمرین
exponential / power function	نمایی / توانی	دو شکل رایج برای منحنی فراموشی و یادگیری

English	فارسی	توضیح کوتاه
sampling distribution	توزیع نمونه‌گیری	توزیع برآورد در تکرارهای فرضی آزمایش
standard error (SE)	خطای استاندارد	پراکندگی برآورد در تکرارهای فرضی آزمایش
confidence interval (CI)	بازه‌ی اطمینان	بازه‌ای که با روش مشخص پارامتر واقعی را در بر می‌گیرد
Hessian	ماتریس Hessian	ماتریس مشتق‌های دوم؛ انحنای درست‌نمایی
log-likelihood / NLL	منفی آن / لگاریتم درست‌نمایی	مقداری که بهینه‌سازها کمینه یا بیشینه می‌کنند
Wald interval	بازه‌ی Wald	برآورد $\pm$ 1.96 SE
profile likelihood	درست‌نمایی نیم‌رخ (profile)	درست‌نمایی یک پارامتر با بهینه کردن بقیه
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها
bootstrap	bootstrap (بازنمونه‌گیری)	تخمین عدم قطعیت با شبیه‌سازی یا بازنمونه‌گیری
parameter trade-off	بده‌بستان پارامترها	تغییر یک پارامتر که با پارامتر دیگر جبران می‌شود
aggregation artifact	خطای ناشی از میانگین‌گیری	الگویی در میانگین که در هیچ فردی نیست
trial	کوشش (trial)	یک بار انجام تکلیف توسط شرکت‌کننده
bounds / constraints	قیدها / کران‌ها	محدودیت‌های مجاز برای مقدار پارامترها
accuracy	دقت	نسبت پاسخ‌های درست
complete / no / partial pooling	ادغام جزئی / بدون ادغام / ادغام کامل (pooling)	/ جدا برای هر فرد / یک پارامتر برای همه / سلسله‌مراتبی
individual differences	تفاوت‌های فردی	فرق پارامترها بین افراد
noise	نویز	تغییرات تصادفی در شواهد یا در داده
benchmark	benchmark (مجموعه‌آزمون استاندارد)	مجموعه سؤال برای مقایسه‌ی مدل‌های هوش مصنوعی

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Explain the sampling distribution of an estimator and distinguish a standard error from a standard deviation of the data.
2. Compute standard errors and confidence intervals for model parameters in three ways: from the Hessian of the NLL (Wald), from the profile likelihood, and by parametric bootstrap. Compare these three methods.
3. Diagnose parameter trade-offs from the estimated correlation between parameters.

4. Demonstrate by simulation that averaging individual curves can produce a functional form (a "power law") that no individual follows.
5. Choose between aggregate, individual and (preview) hierarchical fitting, and justify the choice.

1.1.3 Motivation & big picture

"$\hat{d} = 0.42$" is half a result. Last week the power model's MLE for the Week 2 data was $\hat{d} = 0.42$ (true value 0.5). Is 0.42 compatible with 0.5? With 0.3? With 0.6? Without a measure of uncertainty we cannot say whether two participants, two groups or two conditions really differ. Every parameter estimate we report from now on comes with a confidence interval.

The law that nobody obeys. The *power law of practice* is one of the most cited regularities in cognitive psychology: response times fall with practice approximately as a power function of the number of trials. Some researchers have argued that the power shape can arise from averaging over people (or items) who each improve *exponentially* but at different rates (see Further reading). If so, the "law" describes the average but no single learner, and a theory built to explain a power function in each individual mind would be explaining an artifact. This week we demonstrate the effect with simulated data and learn when to fit individuals rather than group averages.

Where is this used?

- Any claim of a difference ("patients forget faster than controls", "training improved sensitivity") needs uncertainty estimates. Without them a difference of 0.05 in a parameter may be pure noise.
- Clinical assessment of individuals: to say that one patient's memory decay is abnormal, you need the standard error of *that patient's* estimate, not only the group average.
- Learning and skill research: the "power law of practice" was long treated as a universal law. This week's aggregation artifact shows why researchers now fit individual learners (and why the same caution applies to group-averaged curves in education, sports science and human factors).
- Replication and meta-analysis combine estimates from many studies by weighting each one by its precision ($1/\mathrm{SE}^2$).

1.1.4 Lecture notes

In plain words If you ran the same experiment again on the same person, you would get slightly different data and therefore a slightly different parameter estimate. The standard error (SE) says how much the estimate would jump around between such repetitions; a confidence interval is a range built from it (roughly estimate $\pm$ 2 SE). We cannot repeat the experiment many times, so we estimate this jumping in three ways: (1) from how sharply peaked the likelihood is (a sharp peak means the data pin the parameter down; this is the Wald method using the Hessian), (2) by tracing how fast the fit gets worse as we move one parameter (profile likelihood), and (3) by simulating many fake experiments from the fitted model and refitting each (bootstrap). The second half of the week warns about a trap: averaging data across people can create a curve shape that no single person has. If every learner improves exponentially but at a different speed, the group average looks like a power law.

Suggested timing (60 min): sampling distributions 10 · Hessian standard errors 15 · profile likelihood 8 · bootstrap 10 · multiple participants and aggregation 17.

1. Estimates have sampling distributions

If we reran the same experiment on the same participant (same true $\boldsymbol{\theta}^*$), we would get different data and therefore a different $\hat{\boldsymbol{\theta}}$. The distribution of $\hat{\boldsymbol{\theta}}$ over hypothetical replications is its sampling distribution.

- The standard error (SE) of a parameter is the SD of its sampling distribution: how much the *estimate* would vary across replications.
- A 95% confidence interval is constructed by a procedure that, across replications, contains θ^* in 95% of cases. (It is *not* "a 95% probability that θ^* is in this particular interval"; that is a Bayesian credible interval, Week 9.)

We cannot rerun the experiment many times, so we approximate the sampling distribution in three ways.

2. Standard errors from the curvature of the likelihood (Wald)

Intuition. If the NLL has a sharp, narrow minimum, small changes in $\boldsymbol{\theta}$ make the data much less likely, so the data pin $\boldsymbol{\theta}$ down precisely. A flat, wide minimum means many values fit almost equally well, so the uncertainty is large. Curvature is measured by second derivatives.

Formalism. The Hessian of the NLL at the MLE is the matrix of second derivatives,

$$\mathrm{H}_{mn} = \left.\frac{\partial^2 \,\mathrm{NLL}(\boldsymbol{\theta})}{\partial\theta_m \,\partial\theta_n}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}},$$

also called the observed Fisher information. For large samples,

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\theta}}) \approx \mathrm{H}^{-1}, \qquad \mathrm{SE}(\hat{\theta}_m) = \sqrt{[\mathrm{H}^{-1}]_{mm}}, \qquad 95\%\text{ Wald CI: } \hat{\theta}_m \pm 1.96\,\mathrm{SE}(\hat{\theta}_m).$$

The off-diagonal elements of H^{-1}, rescaled to correlations, show parameter trade-offs: a correlation of -0.9 between two estimates means their errors largely compensate each other (the tilted valley of Week 3).

Sanity check (binomial). For one proportion, $\mathrm{NLL}''(\theta) = y/\theta^2 + (n-y)/(1-\theta)^2$. At $\hat{\theta} = y/n$ this equals $n/[\hat{\theta}(1-\hat{\theta})]$, so $\mathrm{SE} = \sqrt{\hat{\theta}(1-\hat{\theta})/n}$, the familiar textbook formula.

In practice: compute H numerically by finite differences (lab Part A).

When Wald fails: small samples, estimates near a bound (e.g., a recall level $\hat{A}$ close to 1), strongly non-quadratic likelihoods, or parameters with skewed sampling distributions (rates, variances). Then the symmetric ± 1.96 SE interval can extend into impossible values.

> Advanced: why this works. Near its minimum, any smooth NLL is approximately quadratic (second-order Taylor expansion). A quadratic NLL is exactly the NLL of a multivariate normal distribution whose inverse covariance is the Hessian. Combined with the asymptotic normality of the MLE (Week 4), this gives the result. The same quadratic approximation, called the *Laplace approximation*, reappears in Bayesian model comparison (Week 10).

Reading the math

- $\mathrm{H}_{mn} = \partial^2 \mathrm{NLL}/\partial\theta_m \partial\theta_n$ at $\hat{\boldsymbol{\theta}}$: the matrix of second derivatives, i.e. the curvature of the NLL bowl at its bottom. Diagonal entries: how steep the bowl is along each parameter. Off-diagonal entries: whether the bowl is tilted (parameters trade off).
- $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\theta}}) \approx \mathrm{H}^{-1}$: a sharply curved bowl (large H) gives a small covariance, i.e. precise estimates. Taking the inverse is what turns "steepness" into "uncertainty".
- $\mathrm{SE}(\hat{\theta}_m) = \sqrt{[\mathrm{H}^{-1}]_{mm}}$: take the m-th diagonal entry of the inverse and its square root.
- One-parameter check: for a proportion, $\mathrm{NLL}'' = n/[\hat{\theta}(1-\hat{\theta})]$, so $\mathrm{SE} = \sqrt{\hat{\theta}(1-\hat{\theta})/n}$. With 30 of 50 recalled: $\sqrt{0.6 \times 0.4/50} = 0.069$, and the 95% interval is $0.60 \pm 1.96 \times 0.069 = [0.46, 0.74]$.
- Correlation from the covariance: $\mathrm{corr} = \mathrm{Cov}_{12}/(\mathrm{SE}_1 \mathrm{SE}_2)$. A value like -0.9 means the two estimates are strongly traded off.

3. Profile likelihood intervals

For a parameter of interest ψ (say d), the profile NLL is obtained by fixing ψ on a grid and minimizing the NLL over all *other* parameters at each grid value:

$$\mathrm{NLL}_{\mathrm{prof}}(\psi) = \min_{\text{other params}} \mathrm{NLL}(\psi, \text{others}).$$

The 95% likelihood-ratio (profile) interval is the set of ψ values that are not much worse than the best:

$$\{\psi : 2\,[\mathrm{NLL}_{\mathrm{prof}}(\psi) - \mathrm{NLL}_{\min}] \le 3.84\}, \qquad 3.84 = \chi^2_{1,\,0.95}.$$

Profile intervals follow the actual shape of the likelihood, so they can be asymmetric and respect natural bounds. They are usually more accurate than Wald intervals in small samples.

Reading the math

- $\mathrm{NLL}_{\mathrm{prof}}(\psi)$: fix the parameter of interest ψ at one value, let the optimizer choose the best values of all *other* parameters, and record the resulting NLL. Repeat for many values of ψ. The result is a 1-D curve.
- The interval keeps every ψ whose profile NLL is within $3.84/2 = 1.92$ units of the minimum. The number 3.84 is the 95% point of a χ^2 distribution with 1 df; it comes from the likelihood-ratio test (Week 7).
- Unlike Wald, the curve is not forced to be a symmetric parabola, so the interval can be lopsided and never extends to impossible values.

4. The parametric bootstrap

A simulation answer to "what if we reran the experiment?":

1. Fit the model to the data: $\hat{\boldsymbol{\theta}}$.
2. Simulate B new datasets from the model at $\hat{\boldsymbol{\theta}}$, with the same design as the real experiment.
3. Refit each dataset: $\hat{\boldsymbol{\theta}}^{*(1)}, \ldots, \hat{\boldsymbol{\theta}}^{*(B)}$.
4. The SD of the bootstrap estimates approximates the SE. The 2.5th and 97.5th percentiles give a percentile CI.

The bootstrap needs no derivatives, works for any statistic (e.g., a half-life, a difference between conditions), and shows the full shape and correlations of the sampling distribution. Its cost is B refits ($B \approx 500$–2000). The non-parametric bootstrap resamples the observed trials with replacement instead of simulating from the model.

Comparison:

Table 1.2: 4. The parametric bootstrap

Method	Cost	Symmetric?	Handles bounds?	Needs
Wald (Hessian)	One Hessian	Always	No	Smooth NLL, large N
Profile likelihood	One fit per grid value	Can be asymmetric	Yes	A likelihood
Parametric bootstrap	B fits	Can be asymmetric	Yes	A model you can simulate

5. Fitting multiple participants

Real experiments have J participants (index j). There are three broad strategies:

Table 1.3: 5. Fitting multiple participants

Strategy	What is fitted	Pros	Cons
Aggregate (complete pooling)	One model to the averaged data	Stable; works with few trials per person	Assumes everyone is the same; can distort the functional form
Individual (no pooling)	One model per participant; then analyze the J sets of estimates	Respects individual differences; standard practice	Noisy estimates with few trials; second-stage analysis ignores estimation uncertainty
Hierarchical (partial pooling)	Individual parameters drawn from a group distribution, estimated jointly	Best of both; principled shrinkage	More complex; usually Bayesian (Week 13)

Aggregation artifacts. The average of nonlinear curves is generally not a curve of the same family. For example, averaging exponential curves with different rates does not give an exponential: fast learners finish early, so the average's relative rate of change *slows down* over time, which is the defining property of a power function (Week 2, relative loss rate).

> Advanced: an exact result. Suppose each participant follows $e^{-\lambda_j t}$ and the rates are Gamma-distributed across people, $\lambda_j \sim$ Gamma(shape s, rate b). Then the group average curve is
>
> $$\mathbb{E}\left[e^{-\lambda t}\right] = \left(1 + \frac{t}{b}\right)^{-s},$$
>
> which is *exactly* a shifted power function (this is the moment-generating function of the Gamma distribution, evaluated at $-t$). Everyone forgets exponentially, yet the average forgets by a power law with exponent equal to the Gamma shape s. You verify this numerically in Exercise 4.

Practical guideline. Fit individuals whenever each participant has enough data to constrain the model; use the aggregate only as a summary, never as evidence about the *form* of an individual-level process. When individual data are sparse, use hierarchical models (Week 13).

For an engineer

- Wald SE from the Hessian = the inverse Fisher information, i.e. the CRLB evaluated at the estimate. MLE reaches the bound asymptotically, so for large N the Wald interval is the CRLB-based interval.

- The parametric bootstrap is the Monte Carlo performance evaluation you do for an estimator or a receiver (e.g. simulating many noise realizations to get an estimator's variance or a BER curve), done at the fitted parameter values.
- Profile likelihood is what you do when you have *nuisance parameters*: optimize them out and look at the likelihood of the parameter you care about.
- Aggregation artifact = the difference between an ensemble average and a single realization. Averaging exponential decays with different time constants gives a slower, heavy-tailed decay, just as the average of many channel realizations does not look like any one realization. The process is not "ergodic across people": the group mean does not describe the individual.
- Partial pooling (preview of Week 13) is like estimating each user's SNR by combining their own few measurements with the distribution of SNRs in the whole cell: an empirical-Bayes estimate.

For the instructor

- Start with the question "is 0.42 different from 0.5?" (Motivation). Students will want a yes/no; show that the answer depends entirely on the SE.
- Show the three methods side by side on the same data (lab Part C table). The message is not "which is correct" but "they agree when the likelihood is well-behaved, and differ near bounds or with little data".
- The aggregation demo is the highlight. Plot 5 individual exponential curves and their average on a log–log plot: the average is almost a straight line (power law), while each individual is curved. Ask the class what a researcher who only saw the average would conclude.
- Typical questions.
  - *"What is the difference between SD and SE?"* SD describes the spread of the *data*; SE describes the spread of an *estimate* across repeated experiments. SE shrinks with more data; SD does not.
  - *"Does a 95% CI contain the true value with probability 0.95?"* Not this particular interval; 95% of intervals built this way would. (The Bayesian credible interval of Week 9 does have the direct reading.)
  - *"Should I always fit individuals?"* Yes, if each person has enough data; otherwise use hierarchical models (Week 13).

1.1.5 Worked example

Part 1: three intervals for a recall probability

A participant recalls $y = 30$ of $n = 50$ items; $\hat{\theta} = 0.60$.

Wald. SE $= \sqrt{0.6 \times 0.4/50} = 0.0693$, so the 95% CI is $0.60 \pm 1.96 \times 0.0693 = [0.464, 0.736]$.

Profile (likelihood ratio). Solve $2[\ell(0.6) - \ell(\theta)] = 3.84$ numerically on both sides of 0.6. The 95% CI is $[0.462, 0.728]$, slightly asymmetric: it extends 0.138 below the estimate and 0.128 above, because the likelihood is skewed toward 0.5, where binomial variance is largest.

Interpretation. With 50 items, a recall probability is only known to about ± 0.13. A difference of 0.10 between two conditions of this size would not be convincing on its own.

Part 2: averaging two exponential learners

Participant 1 forgets at $\lambda = 0.1$, participant 2 at $\lambda = 1.0$. The average curve $\bar{f}(t) = (e^{-0.1t} + e^{-t})/2$:

Table 1.4: Part 2: averaging two exponential learners

t	0	1	2	5	10
$\bar{f}(t)$	1.000	0.636	0.477	0.307	0.184
Relative loss rate $-\bar{f}'(t)/\bar{f}(t)$	0.55	0.36	0.23	0.11	0.10

Each participant has a *constant* relative loss rate (0.1 and 1.0), but the average's rate falls from 0.55 to 0.10, the signature of a power-like curve. The reason: at later times the average is dominated by the slow forgetter, whose rate is low. Interpretation: a researcher fitting only the average would conclude that "memories become more resistant with age", a claim true of neither participant.

1.1.6 Lab (2 hours)

Goals

1. Compute Wald, profile and bootstrap intervals for the power model fitted to the Week 2 data, and compare them.
2. Visualize parameter trade-offs through the estimated correlation and the bootstrap cloud.
3. Simulate 20 exponential learners, fit exponential and power models to the average and to each individual, and interpret the contradiction.

Timing: Part A 20 min · Part B 15 min · Part C 25 min · Part D 40 min · Part E 20 min.

Part A: Wald standard errors from a numerical Hessian (20 min)

Interpret: both 95% intervals contain the true values ($A^* = 0.9$, $d^* = 0.5$), so the "miss" we worried about in Weeks 3–4 is well within sampling error. Note that the Wald interval for A happens to end at 1.000, right at the boundary of possible values. With slightly noisier data, the symmetric Wald interval would include impossible values (probabilities above 1), whereas the bootstrap interval cannot. The positive correlation is the trade-off valley from Week 3: when the estimate of A is too high, the estimate of d tends to be too high as well.

Part C: parametric bootstrap (25 min)

Compare: put the Wald, profile and bootstrap intervals for d side by side. Are they similar? Which is asymmetric? With these data (8 cells × 50 items) all three methods should agree reasonably well. They diverge more with fewer items or with estimates near a bound (Exercise 5).

Part D: the power law of practice as an aggregation artifact (40 min)

Design. $J = 20$ participants each complete 60 practice blocks of a visual-search task. Each participant improves exponentially:

$$\mathrm{RT}_j(m) = E + B\, e^{-r_j (m-1)} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 40\ \mathrm{ms}),$$

with asymptote $E = 400$ ms, initial gain $B = 600$ ms, and individual learning rates $r_j \sim \mathrm{Gamma}(\mathrm{shape} = 0.7, \mathrm{scale} = 0.5)$ (mean 0.35 per block; a skewed distribution, with a few very fast learners and several very slow ones). Here m is the practice block, and ε is the noise of a block-mean RT.

Part E: interpret (20 min)

Write 2–3 sentences for each: 1. Which statement about *learning* is justified, and which is not? 2. Why is the exponential rate fitted to the average curve smaller than the mean of the individual rates? (Hint: which participants dominate the average late in practice?) 3. Individual fits recover the true rates well here, with 60 blocks per person. What would change with only 10 blocks per person, and what would you do then? (Preview of Week 13.)

Exercises (increasing difficulty)

Exercise 1 ⋆. Compute Wald 95% CIs for the *exponential* model's MLE on the Week 2 data. Do they contain sensible values? What happens to the Hessian at a point that is not a minimum?

Exercise 2 ⋆. Rerun the bootstrap with $B = 50, 200, 1000$ and different seeds. How stable are the percentile limits? How large should B be for a 95% CI?

Exercise 3 ⋆⋆ (a derived quantity). Use the bootstrap samples from Part C to compute a 95% CI for the half-life of the power model, $t_{1/2} = 2^{1/d} - 1$. Why is this easy with the bootstrap and awkward with the Hessian?

Exercise 4 ⋆⋆ (the Gamma–power result). Draw 100 000 rates $\lambda \sim$ Gamma(shape $s = 0.8$, rate $b = 2$), average $e^{-\lambda t}$ over them for $t \in [0, 30]$, and overlay the analytic curve $(1 + t/b)^{-s}$ (Lecture §5, Advanced box). Then fit exponential and power functions to the averaged curve.

Exercise 5 ⋆⋆⋆ (Advanced: coverage). How often do 95% intervals actually contain the true value? Simulate 300 datasets from the power model ($A = 0.9$, $d = 0.5$) with $n = 10$ items per interval, and compute the Wald and profile CIs for d for each. Report the coverage (proportion containing 0.5) of each method. Which is closer to 95%? Repeat with $n = 50$.

Lab check-off

Show: (1) a table comparing the Wald, profile and bootstrap 95% CIs for d; (2) the bootstrap scatter plot with one sentence on the trade-off; (3) the individual-vs-average comparison and your answer to interpretation question 1.

1.1.7 Connection to AI & LLMs

Benchmark scores have standard errors. AI models are compared on benchmarks, and leaderboards often rank models that differ by one or two percentage points. A benchmark is a *sample* of items, so its accuracy has sampling error just like a parameter estimate. Two lessons from this week apply directly: 1. Report uncertainty. A bootstrap over items gives a confidence interval for the accuracy and for the difference between two models. 2. Respect the design. Both models answered the *same* items, so the comparison is paired, like a within-participant design. Resampling the two models independently ignores the shared item difficulty and gives intervals that are too wide.

Item difficulty also varies a lot, which is the item-level version of the participant heterogeneity in Part D: a single average accuracy hides it.

A 7-point lead on 100 items is not clearly different from zero. A leaderboard that ranks A above B without an interval is overconfident.

Find out by simulation. - ⋆⋆ Compute the Wald SE of each accuracy, $\sqrt{p(1-p)/n}$, and compare it with the bootstrap SE. - ⋆⋆ Discussion: a paper reports “Model A: 74%, Model B: 67%” with no interval and concludes that A “reasons better”. List what you would ask the authors.

1.1.8 Common mistakes & misconceptions

1. Confusing SE with SD. The SD describes the spread of the *data*; the SE describes the uncertainty of an *estimate* and shrinks as the data grow.
2. “There is a 95% probability that the true value is in my CI.” For a frequentist CI, 95% refers to the long-run behavior of the procedure. The probability statement belongs to Bayesian credible intervals (Week 9).

3. Compute the Hessian properly.
4. Trusting Wald intervals near bounds. If $\hat{\theta}$ is on or near a bound, or the interval crosses impossible values, use profile or bootstrap intervals.
5. Reporting parameters without uncertainty. Comparing two point estimates ("participant 1 has $d = 0.42$, participant 2 has $d = 0.51$") is meaningless without intervals.
6. Inferring individual-level processes from averaged data. Averages of nonlinear curves change functional form (Part D). This applies to items, trials and participants alike.
7. Treating individual estimates as if they were data. Second-stage t-tests on fitted parameters ignore that some estimates are much noisier than others. Hierarchical models fix this (Week 13).

1.1.9 Check your understanding

1. What is the sampling distribution of an estimator, and why can't we observe it directly? Answer The distribution of the estimates we would get across many hypothetical replications of the experiment with the same true parameters. We have only one dataset, so we approximate it with the Hessian (asymptotic theory), the profile likelihood, or simulation (bootstrap).

2. The NLL has a very flat minimum along one direction of the parameter space. What does that imply for the standard errors and the correlation of the estimates? Answer Low curvature means a small Hessian entry in that direction, so a large variance in H^{-1}: large SEs. If the flat direction is diagonal, the estimates are strongly correlated (a trade-off).

3. Derive the Wald SE for a binomial proportion from the second derivative of the NLL. Answer $\mathrm{NLL}''(\theta) = y/\theta^2 + (n-y)/(1-\theta)^2$. At $\hat{\theta} = y/n$ this is $n/\hat{\theta} + n/(1-\hat{\theta}) = n/[\hat{\theta}(1-\hat{\theta})]$. Inverting and taking the square root gives $\mathrm{SE} = \sqrt{\hat{\theta}(1-\hat{\theta})/n}$.

4. List the steps of a parametric bootstrap for a CI on a model parameter. Answer Fit the model to get $\hat{\boldsymbol{\theta}}$. Simulate B datasets from the model at $\hat{\boldsymbol{\theta}}$ using the original design. Refit each one. Use the SD of the refitted estimates as the SE, and their 2.5th and 97.5th percentiles as the 95% CI.

5. Why can a profile-likelihood interval be asymmetric while a Wald interval cannot? Answer The Wald interval assumes a quadratic NLL (a normal sampling distribution), which is symmetric by construction. The profile interval uses the actual likelihood, which can be skewed, e.g., near a bound or for rate parameters.

6. Twenty participants each learn exponentially, with different rates. The averaged curve is fitted better by a power function. Should we conclude that learning follows a power law? Answer No. The power shape is an aggregation artifact: averaging exponentials with heterogeneous rates produces a curve whose relative rate of change decreases over time. Conclusions about the form of individual learning must come from individual (or hierarchical) fits.

7. Give one advantage and one disadvantage of fitting each participant separately. Answer Advantage: respects individual differences and avoids aggregation artifacts. Disadvantage: with few trials per person, estimates are noisy (and sometimes extreme or on bounds), and analyzing them in a second stage ignores their different uncertainties.

1.1.10 Assignment

No new graded assignment this week. A2 (LS vs. MLE) is due at the start of the Week 6 lecture. Part A of today's lab shows how to add standard errors to your A2 estimates. Doing so is not required, but it will strengthen your interpretation.

Optional practice (ungraded): Exercise 5 (coverage) is the best preparation for Week 7's recovery studies.

1.1.11 Further reading

- Farrell & Lewandowsky (2018), chapter 5. Combining information from multiple participants: aggregate vs. individual fitting and the dangers of averaging. *(Main reading.)*
- Farrell & Lewandowsky (2018), chapter 4 (the sections on parameter uncertainty and the Hessian), revisited from last week.
- *Optional, outside the course list:* Heathcote, A., Brown, S., & Mewhort, D. J. K. (2000). The power law repealed: The case for an exponential law of practice. *Psychonomic Bulletin & Review, 7*, 185–207. The classic analysis of averaging artifacts in practice data.

Preview for Week 6: Exercise 4 of Week 3 showed a three-parameter model beating the true model on SSE. Next week we confront this directly: why good fit is not enough, and how cross-validation measures what we really want, which is prediction of new data.